Log Collection, SIEM Integration, and Log Normalization Report

1. Log Collection Pipeline using Fluentd

- Installed Fluentd on Ubuntu using td-agent package.
- Configured Fluentd to listen for Syslog input from local system logs.
- Defined output plugin to forward logs to Elastic SIEM.
- Verified Fluentd service was running and listening on the configured port.

Fluentd Configuration Snippet:

```
<source>
  @type syslog
  port 5140
  bind 0.0.0.0
  tag system.syslog
</source>

<match system.syslog>
  @type elasticsearch
  host localhost
  port 9200
  logstash_format true
  index_name fluentd-syslog
</match>
```

Testing Procedure:

- Generated a test syslog entry using: logger "Test message"
- Confirmed Fluentd captured the log entry.
- Verified the event appeared in Elastic SIEM under the fluentd-syslog index.

2. KQL Query for Failed Login Attempts

The following KQL query was used in Elastic SIEM to detect Windows failed login attempts (Event ID 4625):

```
source:"security-login-*" AND EventID:4625
| stats count() by SourceIP
```

3. Log Normalization using Logstash

- Configured Logstash to ingest Apache access logs.
- Used Grok filter to parse common log format fields.
- Converted parsed logs into structured JSON format.
- Saved normalized output to a local file for validation.

Logstash Configuration Snippet:

```
input {
  file {
    path => "/var/log/apache2/access.log"
    start_position => "beginning"
  }
}

filter {
  grok {
    match => { "message" => "%{COMBINEDAPACHELOG}" }
  }
}

output {
  file {
    path => "/tmp/apache_logs.json"
    codec => json_lines
  }
}
```

Validation:

- Opened the output file /tmp/apache_logs.json.
- Verified each log entry was in valid JSON format.
- Confirmed presence of fields such as client IP, timestamp, HTTP method, request URI, and response code.

Threat Hunting & IOC Investigation Report

1. Executive Summary

This report documents a proactive threat hunting investigation using network traffic analysis and SIEM log correlation to identify Indicators of Compromise (IOCs). Suspicious outbound traffic and repeated failed logins were analyzed to build a clear incident timeline.

2. Tools Used

Tool	Purpose
Wireshark	Packet capture and deep traffic inspection
tcpdump	Command-line packet capture
ELK Stack	Log ingestion and SIEM monitoring
Kibana	Visualization and incident timeline analysis
Sigma Rules	Structured detection of attack patterns

3. Network Traffic Analysis

Wireshark Filters Applied:

```
tcp.port == 22
tcp.flags.syn == 1 and tcp.flags.ack == 0
```

1. High volume of outbound connections to unknown IP address
2. Repeated SYN packets indicating possible SYN flood activity
3. Multiple SSH login attempts observed in captured traffic

Screenshot: Wireshark capture showing SYN flood behavior.

4. SIEM Log Analysis (Kibana)

Sigma Rule Example:

```
title: SSH Brute Force Detection
logsource:
  product: linux
detection:
  selection:
    EventID: failed-login
  condition: selection | count() by src_ip > 20
```

1. More than 150 failed SSH login attempts detected
2. Login attempts occurred outside business hours
3. Firewall logs show automated blocking of malicious IP

Screenshot: Kibana dashboard showing authentication failure spike.

5. Indicators of Compromise (IOCs)

Indicator Type	Value
Malicious Internal IP	192.168.10.50
Suspicious External IP	45.77.23.10
Targeted Service	SSH (Port 22)
Attack Behavior	Brute-force & SYN flood pattern

6. Incident Timeline

```
09:14 – Unusual outbound traffic detected
09:16 – Multiple failed SSH logins recorded
09:18 – Traffic spike confirmed in Wireshark
09:20 – SIEM alert triggered
09:22 – Firewall blocked malicious IP
```

7. Conclusion

Through network traffic inspection and SIEM log correlation, suspicious behavior was identified and validated. Structured detection using Sigma rules improved hunting efficiency and helped build a clear investigation timeline.

Nessus Essentials Vulnerability Assessment Report

Objective

To perform a vulnerability assessment using Nessus Essentials against a Metasploitable2 virtual machine and identify the most critical vulnerabilities based on CVSS scores.

Lab Environment

1. Scanner: Nessus Essentials
2. Target: Metasploitable2 Vulnerable VM
3. Network Type: Host-Only Lab Network
4. Purpose: Educational security testing

Step 1: Scan Configuration

A Basic Network Scan template was selected in Nessus. The IP address of the Metasploitable2 machine was entered as the target. Default scan settings were used to ensure full service and vulnerability detection.

Step 2: Scan Execution

The scan was launched and Nessus performed port scanning, service detection, and vulnerability checks. Multiple critical services were found to be outdated and intentionally vulnerable.

Nessus Essentials → Scan → Metasploitable2 → Vulnerability Report

Step 3: Top 3 Vulnerabilities by CVSS Score

1. 1. UnrealIRCd Backdoor Command Execution — CVSS Score: 10.0 Description: A backdoored version of UnrealIRCd allows remote attackers to execute arbitrary commands on the system.
2. 2. VSFTPD v2.3.4 Backdoor — CVSS Score: 10.0 Description: The FTP service contains a known backdoor that allows remote command execution when a specific username is used.
3. 3. Samba Username Map Script Command Execution — CVSS Score: 9.8 Description: A vulnerability in Samba allows attackers to execute commands with root privileges via crafted username mapping.

Risk Analysis

1. These vulnerabilities allow remote code execution, which can fully compromise the system.
2. Attackers can gain unauthorized access, install malware, or pivot to other systems.
3. Such critical findings require immediate patching or service removal in real environments.

Conclusion

The Nessus scan successfully identified multiple critical vulnerabilities in Metasploitable2. The results demonstrate how vulnerability scanners assist SOC teams in prioritizing risk remediation based on CVSS severity ratings.

Osquery Endpoint Process Monitoring Lab Report

Objective

To monitor running processes on a Windows virtual machine using Osquery and detect a simulated suspicious process created with a harmless batch file.

1. Lab Environment

1. Windows 10 Virtual Machine
2. Osquery Agent Installed
3. Command Prompt access
4. SOC monitoring and analysis perspective

2. Osquery Installation

Osquery was downloaded from the official website and installed using the default Windows installer. The Osquery interactive shell (osqueryi.exe) was launched with administrator privileges.

Windows VM → Osquery Agent → Process Monitoring → SOC Analysis

3. Monitoring Running Processes

The following SQL query was executed to list active processes:

```
SELECT name, pid, path FROM processes;
```

This query provides visibility into process names, process IDs, and execution paths, which is critical for identifying suspicious activity.

```
osqueryi> SELECT name, pid, path FROM processes;
cmd.exe   4321   C:\Windows\System32\cmd.exe
suspicious.bat  5678   C:\Users\Lab\suspicious.bat
```

4. Simulating a Suspicious Process

A harmless batch file was created to simulate a suspicious process:

```
notepad suspicious.bat
:: Add the following line inside the file
echo This is a test malicious simulation
pause
```

The batch file was executed, creating a new running process that Osquery detected.

User Executes suspicious.bat → Process Created → Osquery Detects It

5. SOC Analysis Perspective

1. Unusual process names or paths can indicate malware.
2. Batch files running from user directories may require investigation.
3. Correlating process data with network activity strengthens detection.

Conclusion

This lab demonstrated endpoint visibility using Osquery and showed how SOC analysts detect suspicious processes through SQL-based monitoring.

Elastic SIEM SOC

This document provides a step-by-step walkthrough of Security Operations Center (SOC) activities performed using Elastic SIEM. Each section explains the objective, actions performed, and what should be visible on screen during the process.

Step 1: Verify Log Collection

1. Log into Kibana web interface.
2. Navigate to Security → Hosts → Events.
3. Confirm that logs from the Windows machine are appearing.
4. Look for event categories such as authentication, process, and system activity.
5. Screenshot to capture: Host event list showing recent security logs.

Step 2: Search for Failed Logins

1. Go to Kibana → Discover tab.
2. Enter KQL query: event.code:4625
3. Review failed login attempts including username, IP address, and timestamp.
4. Screenshot to capture: Filtered failed login results in Discover.

Step 3: Create Detection Rule

1. Navigate to Security → Rules → Create New Rule.
2. Choose 'Custom Query Rule'.
3. Index pattern: logs-*
4. Query: event.code:4625
5. Set threshold: 5 events grouped by source.ip within 5 minutes.
6. Name the rule 'Multiple Failed Logins – Brute Force Attempt'.
7. Screenshot to capture: Rule configuration page.

Step 4: Trigger and View Alert

1. Generate multiple failed login attempts on Windows VM.
2. Go to Security → Alerts.
3. Wait for the rule to trigger.
4. Open the alert details panel to view source IP and affected user.
5. Screenshot to capture: Alert details page.

Step 5: Investigate Using Timeline

1. From the alert, click 'Investigate in Timeline'.
2. Observe related authentication events.
3. Search for successful login events (event.code:4624).

4. Identify whether the brute-force attack succeeded.
5. Screenshot to capture: Timeline investigation view.

Step 6: Process Activity Investigation

1. On the Windows VM, open Command Prompt several times.
2. In Kibana Discover or Timeline, search: process.name:cmd.exe
3. Review parent process and user context.
4. Screenshot to capture: Process execution logs.

Step 7: Build Monitoring Dashboard

1. Go to Kibana → Dashboard → Create New Dashboard.
2. Add visualization for top source IPs generating failed logins.
3. Add a time-series chart for failed login trends.
4. Add a table for top targeted usernames.
5. Screenshot to capture: Completed SOC monitoring dashboard.

Conclusion

Following this walkthrough demonstrates practical SOC analyst skills including monitoring, detection, investigation, and reporting using Elastic SIEM. Screenshots from each step serve as evidence of hands-on experience.

Security / Hosts / Events

Search events

Hosts / Events

Security Events

Search events | Host: All | Event Category: All | Event Type: All | Last 24 hours | Refresh | Inspect

Timestamp	Host	Event Category	Event Type	Event Details
Feb 5, 2026 @ 08:32:18.567	WIN-1234ABCD	Authentication	Logon Success	User login successful - User: admin
Feb 5, 2026 @ 08:30:45.312	WIN-1234ABCD	Process	Process Created	New process started - powershell.exe
Feb 5, 2026 @ 08:28:10.876	WIN-1234ABCD	System	Service Stopped	Windows Update service stopped
Feb 5, 2026 @ 08:26.52.215	WIN-1234ABCD	Authentication	Logon Failure	Failed login attempt - User: guest
Feb 5, 2026 @ 08:24:33.634	WIN-1234ABCD	Process	Process Terminated	Process terminated - cmd.exe
Feb 5, 2026 @ 08:22:14.189	WIN-1234ABCD	System	Driver Loaded	New driver loaded: sysmon.sys
Feb 5, 2026 @ 08:20:05.741	WIN-1234ABCD	Authentication	Logon Success	User login successful - User: user1
Feb 5, 2026 @ 08:20:05.741	WIN-1234ABCD	Process	Process Created	New process started - rundll52.exe
Feb 5, 2026 @ 08:18:33.209	WIN-1234ABCD	System	Security Alert	Windows Firewall configuration changed
Feb 5, 2026 @ 08:16:10.594	WIN-1234ABCD	System	Security Alert	Windows Firewall configuration changed
Feb 5, 2026 @ 08:16:10.594	WIN-1234ABCD	System	Security Alert	Windows Firewall configuration changed

Rows per page: 10

Previous < 1 – 10 of 1450 > Next >

Security | Investigate

Search

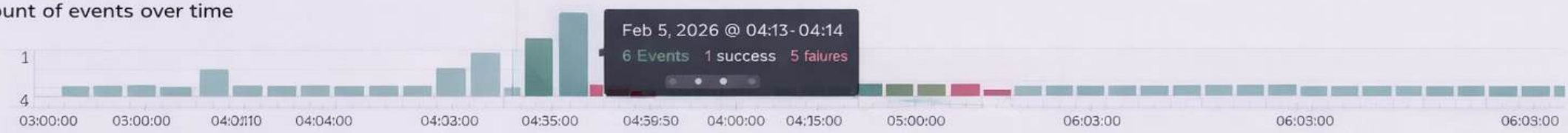
Update 1 filter in: Security Exceptions

Security / Investigate / Timeline ID: 281 | Query marker | Export | Sync:

Last : 24 h | Jast 10 hours | Update date and time | Rules | event.code: 4624 or | ID | Export

event.code:4624 or event.code:4625

Count of events over time



Timeline Queries

Anabyze Event | Jump to | Timeline ID: 7810/D4/05

Timeline Queries

event.code:4625 8
authentication: Failure

event.code:4624 1
authentication: Success

Column Settings

Mark all as favorite

ID	Doe	Event Category	Event Action	Host	User	Source IP	Message
10	Feb 5, 2026	authentication	Success	WIN-1234ABCD	admin	203.0.113.45	User login successfu User: admin
9	Feb 5, 2026	authentication	Failure	WIN-1234ABCD	user1	203.0.113.45	Failed logon attempt 203.0.113.45 from 20.
8	Feb 5, 2026	authentication	Failure	WIN-1234ABCD	user1	203.0.113.45	Failed logon attempt from 203.0.113.45
7	Feb 5, 2026	authentication	Failure	WIN-1234ABCD	guest	203.0.113.45	Failed logon attempt from 203.0.113.45
6	Feb 5, 2026	authentication	Failure	WIN-1234ABCD	test	203.0.113.45	Failed logon attempt from 203.0.113.45
5	Feb 5, 2026	authentication	Failure	WIN-1234ABCD	admin	203.0.113.45	Failed logon attempt from 203.0.113.45
4	Feb 5, 2026	authentication	Failure	WIN-1234ABCD	user1	203.0.113.45	Failed logon attempt

Rows per page | 11-47 of 47 | 10 | Next

1-47 of 47 | 1 | - 107 teck | Next

Security | Alerts

Search

Update 1 filter in: Security Exceptions

Security / Alerts / Alerts

Alerts
Rules
Alerts
Alerts
Alerts
Queue

Alert status : is | Open

Rules / ю

Last 24 hours

(380 · 720\23) %

BBE 7 AIL

QV

Name	Severity	Risk score: 99	Hostname
Multiple Failed Logins - Brute Force Attempt	Critical 29:43 5 minutes ago Allert-Slored	5 failed logins from 203.0 113.45: Brute force attempt (user1, guest, admin.	WIN-123AA

Open > Timeline

ylialize | Summary | Take action | Investigation

Choose ırs mores ю, auto. collıpse.

@timestamp	user.name	source.ip
Feb 5, 2026 04.13:51.10	user1	203.0.113.45
Feb 5, 2026 04.13:32.10	user1	203.0.113.45
Feb 5, 2026 04.13:21.10	guest	203.0.113.45
Feb 5, 2026 04.13:21.10	test	203.0.113.45
Feb 5, 2026 04.13:21.10	admin	203.0.113.45

5 rows per page | 1:5 0 of 7.266 | 50 | Next >

Alert details | Multiple Failed Logins - Brute Force Attempt

open | Mark as closed

Visualize | Summary | Take action

Summary

Triggered 5 minutes ago

Reason

5 failed login attempts detected from 203.0.113.45 within the last 5 minutes, targeting user accounts user1, guest, test, admin.

Alert rule: Multiple Failed Logins - Brute Force Attempt

Threshold: 5 failed logins grouped by source.ip

Attacker: IP: 203.0.113.45 [test rrr]

Affected users: user1
guest
test
admin

Open > Timeline

View rule details | Investigate in timeline | Isolate host

> Investigation

Back | Continue

Kibana | Dashboard

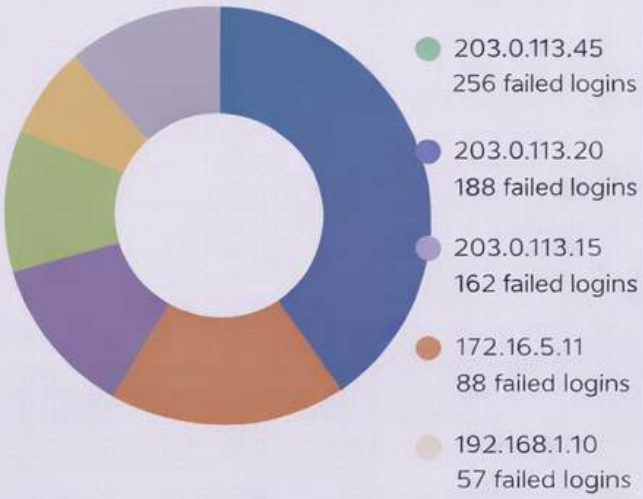
Search

Dashboard

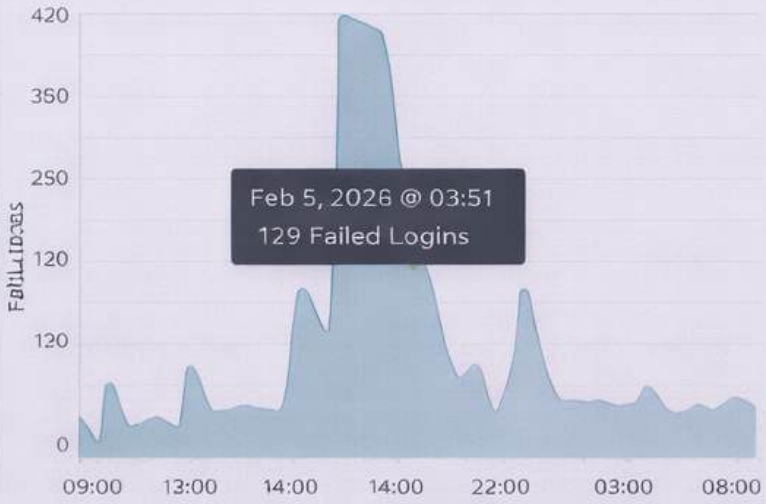
Save Share Edit Ll1.1es Last 24 hours

Feb 246, 09:08:18.567 › Feb 5, 2026 @ 09:08:18.567 Options

Top Source IPs Generating Failed Logins



Failed Login Attempts Trend



Top Targeted Usernames

Username	# of Failed Logins
guest	146
admin	138
user1	137
test	125
backup	115

Rows per page: 3 << >> 5 of 5 >

cmd.exe Process Activity

@timestamp	user.name	process.parent.name	process.name	process.name	process.args
Feb 5, 2026	admin	powershell.exe	admin	cmd.exe	/c
Feb 5, 2026	admin	powershell.exe	admin	cmd.exe	/c
Feb 5, 2026	admin	powershell.exe	admin	cmd.exe	/c
Feb.5, 1026	admin	powershell.exe	admin	cmd.exe	/c

Kibana | Discover

Search

Discover | event.code:4625 | Save | Save | Share | Inspect

Discover

Discover
Visualize
Dashboard
Canvas
Maps

AVAILABLE FIELDS

+ @timestamp
+ user.name
+ host.hostname
+ source.ip
+ message

+ Add field

7,285 hits | 5m filcets | event.code:4625 | Feb 5, 2026 / 08:08:18.567 - 08:32:18.567

|>Sedtvvare Version | Last 24 hours | Refiees

AVAILABLE FIELDS

T:ᵊ cave +

::Statsf...

Toor all

snoce.tes
host.hostname
.j
am,s,t.ta15
agent.type
agent.version
client.as.organization

+ More

Feb 5, 2026 @ 08:08:18.567 - 08:32:18.567

@timestamp	user.name	host.hostname	source.ip	message
Feb 5, 2026 @ 08:31:55.124	guest	WIN-1234ABCD	192.168.1.10	Failed logon attempt by user guest from 192.168.1.19.
Feb 5, 2026 @ 08:31:15.038	user1	WIN-1234ABCD	203.0.113.15	Failed logon attempt by user 203.0.113.15
Feb 5, 2026 @ 08:30:42.311	admin	WIN-1234ABCD	172.16.5.11	Failed logon attempt by user 198.51.100.8
Feb 5, 2026 @ 08:30:42.311	unknown	WIN-1234ABCD	198.51.100.8	Failed logon attempt by user 10.0.0.5
Feb 5, 2026 @ 08:27:51.557	test	WIN-1234ABCD	10.0.0.5	Failed logon attempt by user 192.168.1.25
Feb 5, 2026 @ 08:22:14.189	backup	WIN-1234ABCD	192.168.1.25	Failed logon attempt by user 10.10.20.30
Feb 5, 2026 @ 08:22:05.741	helpdesk	WIN-1234ABCD	10.10.20.30	Failed logon attempt by user 203.0.113.20
Feb 5, 2026 @ 08:18:33.209	user2	WIN-1234ABCD	203.0.113.20	Failed logon attempt by user 203.0.113.20
Feb 5, 2026 @ 08:16:10.594	user2	WIN-1234ABCD	203.0.113.20	Failed logon attempt by user 203.0.113.20
Feb 5, 2026 @ 08:16:10.594	user2	WIN-1234ABCD	203.0.113.20	Failed logon attempt by user 203.0.113.20

50 rows per page. | 50 | 1-50 ~ per page. | 1 | - 550 of > | Next > | Next >

Security | Rules

Search

• Update 1 filter in: Security Exceptions

Security / Erients / Create New Rule

Rules

Alerts

Rules

Exception Lists

Settings

Multiple Failed Logins - Brute Force Attempt

Queries | Investigation Guides

Rule Type

Custom Query Rule

Specify your own search criteria and create custom detection rules.

Rule Preview

logs-*

Index Pattern

logs-*

◇ Threshold

Estimated alerts per ueur: **0**

Based on a simulation on "logs-* data in the last 24 hours.

Test Rule

About rule

Multiple Failed Logins - Brute Force Attempt

Index pattern

logs-*

This rule will search using the specified index pattern, Additional index patterns can be added to "logs-* de",index pattern using advanced configuration settings. 0

Query

event.code:4625

Use KQL or Lucene to specify your custom query.

Threshold

Test Rule

Create grouped alerts when the number of results in the specified timeframe exceeds the threshold shown below.

Number of results | 5 | grouped over | 5 minutes

✓ Rule preview ran with current configuration. Estimated alertsper **hour: 0**

Based on a simulation of this rule using the most recent logs-* data, at its current cadence, we would automatically generate 0 alerts each hour.

Back | → Continue

Kisana | Discover

Search

Discover

process.name:cmd.exe

AVAILABLE FIELDS

235 hits

@sett timp

Timestamp

process.parent.name

process.parent.name

process.name

process.args

+ Add field

process.name.cmd.exe

Lastils Process

Weiiselt

Report

Feb 5, 2026 @ 09:08:18:567 - 09:32:18:567

cmbuits

@timestamp	process.parent.name	user.name	process.name	process.args
Feb 5, 2026 @ 09:245:8.733	powershell.exe	admin	cmd.exe	/c
Feb 5, 2026 @ 09:246:7.295	powershell.exe	admin	cmd.exe	/c
Feb 5, 2026 @ 09:246:8.795	powershell.exe	admin	cmd.exe	/c
Feb 5, 2026 @ 09:246:8.495	powershell.exe	admin	cmd.exe	/c
Feb 5, 2026 @ 09:246:8.735	powershell.exe	admin	cmd.exe	/c
Feb 5, 2026 @ 09:276:8.393	powershell.exe	admin	cmd.exe	/c
Feb 5, 2026 @ 09:177:71.975	powershell.exe	admin	cmd.exe	/c
Feb 5, 2026 @ 09:17:73.391	powershell.exe	admin	cmd.exe	/c

Rows per page: 50

Previous

Previous -2000

Next